Prices, Technical Indicators and Available Information

Calculate each indicator from actual values, then ask which observation or trading condition it supports.

Issues	Questions addressed
1-9: Averages and momentum	Which prices SMA, EMA and MACD use; how their conditions differ
10-16: Size of price changes	Volatility, TR, ATR and stop-loss distance
17-21：RSI	Complete calculation, daily smoothing and overbought conditions
22-25: Volume and signals	Volume, moving-average crossovers and combined conditions
26-27: Calculation timing	Availability of final Closes and centered averages

Issue 5 illustrates one assumed EMA-rule trade, filled at the next Close; it does not rank indicators or tune parameters. Week 4 develops the account, Week 6 adds costs, and Week 7 evaluates the full results.

Which AAPL prices do we share?

AAPL is Apple's ticker. Use the same price sample throughout rather than changing samples between questions.

Predict: Which three dates belong in the three-price average on 1/4?

```
import numpy as np
import pandas as pd
import yfinance as yf
aapl = yf.Ticker("AAPL").history(start="2024-01-02",
    end="2024-07-01", interval="1d", auto_adjust=False, actions=True)
aapl.index = aapl.index.strftime("%Y-%m-%d")
close = aapl["Close"]
pd.DataFrame({'Observation date':close.index[:5],
    'Close (USD/share)':close.iloc[:5].map("{:.2f}".format).values})
```

Observation date	Close (USD/share)
2024-01-02	185.64
2024-01-03	184.25
2024-01-04	181.91
2024-01-05	181.18
2024-01-08	185.56

The 1/2, 3, 4 Closes, 185.64, 184.25, 181.91, are the three inputs in Issue 1. The main sample has 124 observations through 6/28; 7/1 is excluded.

The source Close is split-adjusted, not dividend-adjusted. This week's returns exclude dividends, fees and tax. The short windows, 12/26/9 and Wilder14 are declared teaching settings, not optimal parameters.

Run the code in order; later issues use close and other objects created earlier. Figures use the same data; plotting code is hidden. Reproduction uses retained prices; a new download may include provider revisions.

ISSUE 01

What does an SMA of three Closes show?

A simple moving average (SMA) averages recent prices for comparison; it does not predict a future price.

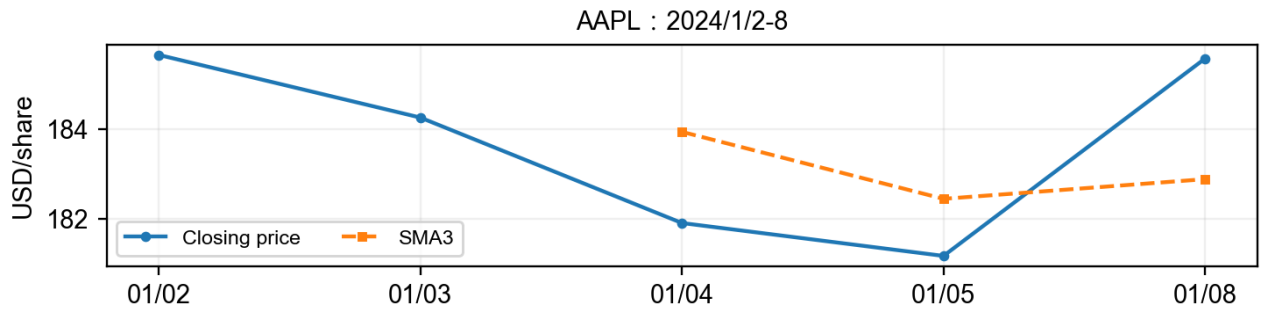
C is Close (USD/share). SMA3 uses today's and the previous two Closes. The first complete window is on 1/4; missing averages on 1/2, 3 must not be filled with zero.

$$\mathrm{SMA3}_t = \frac{C_t + C_{t-1} + C_{t-2}}{3}, \qquad \frac{185.64 + 184.25 + 181.91}{3} \approx 183.9333$$

Predict: On which date, 1/5 or 1/8, does Close cross from below to above the moving average?

```
sma3 = close.rolling(3, min_periods=3).mean()
dates = ["2024-01-04","2024-01-05","2024-01-08"]
pd.DataFrame({'Observation date':dates,'Close (USD/share)':close.loc[dates].round(2),
    'SMA3 (USD/share)':sma3.loc[dates].round(4)})
```

Observation date	Close (USD/share)	SMA3 (USD/share)
2024-01-04	181.91	183.9333
2024-01-05	181.18	182.4467
2024-01-08	185.56	182.8833



The 1/5 Close, 181.18, is below SMA3 182.4467; on 1/8, 185.56 is above 182.8833: an upward crossover.

An upward price crossover of the SMA is a common entry condition. The data are available after the close; meeting the condition does not mean a trade has filled.

Discussion

Which dates supply the prices for SMA3 on 1/5?

Sources: [3], [10].

ISSUE 02

Why can the moving average rise when Close falls?

A moving average changes with the entering and leaving prices, not just yesterday's and today's prices.

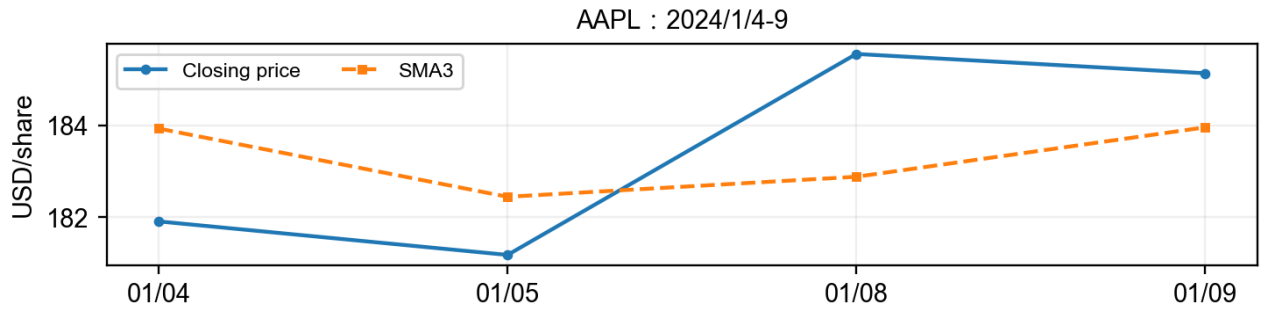
C is Close (USD/share). The 1/8 window uses 1/4, 5, 8; on 1/9 it uses 1/5, 8, 9. The shared 1/5 and 1/8 terms cancel.

$$\mathrm{SMA3}_9 - \mathrm{SMA3}_8 = \frac{C_9 - C_4}{3} \approx \frac{185.14 - 181.91}{3} \approx 1.0767$$

Predict: Close on 1/9 is 0.42 USD below 1/8. Must SMA3 also fall?

```
dates = ["2024-01-04","2024-01-05","2024-01-08","2024-01-09"]
pd.DataFrame({'Observation date':dates,'Close (USD/share)':close.loc[dates].round(2),
    'SMA3 (USD/share)':sma3.loc[dates].round(4)})
```

Observation date	Close (USD/share)	SMA3 (USD/share)
2024-01-04	181.91	183.9333
2024-01-05	181.18	182.4467
2024-01-08	185.56	182.8833
2024-01-09	185.14	183.96



Close falls by 0.42 on 1/9, but adding 185.14 and removing 181.91 raises SMA3 by 1.0767 USD/share.

A rule that considers holding only when the SMA rises qualifies on 1/9. It differs from requiring a same-day price increase.

Discussion

Which entering and leaving prices make the moving average rise on 1/9?

Sources: [3], [10].

ISSUE 03

How does EMA combine today's price with yesterday's average?

An exponential moving average (EMA) retains part of the previous average and adds today's Close.

C and E are Close and EMA (USD/share); α weights today's price. With span=3, α=2/(3+1)=0.5. Initialize E with the 1/2 Close, 185.64, then update it from the previous E each day:

$$E_t = \alpha C_t + (1-\alpha)E_{t-1}, \qquad \alpha = \frac{2}{\text{span}+1}$$

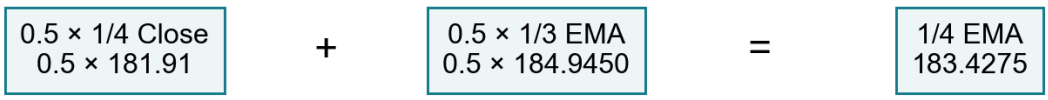
$$E_{\text{Jan3}} = 0.5(184.25) + 0.5(185.64) = 184.945$$

$$E_{\text{Jan4}} = 0.5(181.91) + 0.5(184.945) = 183.4275$$

Predict: Will the 1/4 EMA lie between 181.91 and 184.945?

```
ema3 = close.ewm(span=3, adjust=False).mean()
dates = ["2024-01-02","2024-01-03","2024-01-04"]
pd.DataFrame({'Observation date':dates,'Close (USD/share)':close.loc[dates].round(2),
    'EMA3 (USD/share)':ema3.loc[dates].round(4)})
```

Observation date	Close (USD/share)	EMA3 (USD/share)
2024-01-02	185.64	185.64
2024-01-03	184.25	184.945
2024-01-04	181.91	183.4275



Add the weighted amounts; both are USD/share

The 1/3 value is 184.945; use it as the previous average on 1/4. adjust=False uses this recursion.

Discussion

How much lower is EMA on 1/4 than on 1/3? Use the update equation: below which value must today's Close fall for EMA3 to decrease?

Sources: [4], [11].

ISSUE 04

Why does EMA3 retain prices older than three observations?

EMA is not limited to the last three prices. Older prices gradually lose influence; they do not disappear at the fourth observation.

Keep α=0.5 and first-price initialization. The first price receives the remaining weight, so the first two prices have equal weights. Substitute the 1/3 update into 1/4; E is EMA and C is Close (USD/share):

$$E_{\text{Jan4}} = 0.5C_{\text{Jan4}} + 0.25C_{\text{Jan3}} + 0.25C_{\text{Jan2}}$$

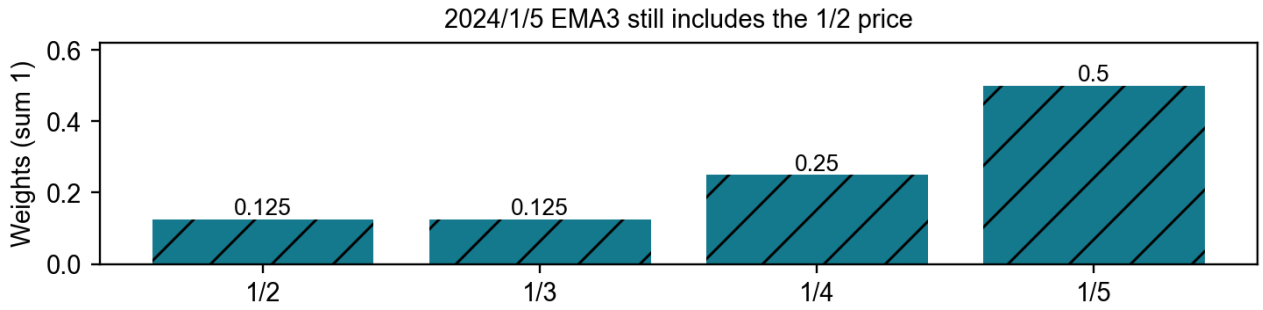
$$E_{\text{Jan5}} = 0.5C_{\text{Jan5}} + 0.25C_{\text{Jan4}} + 0.125C_{\text{Jan3}} + 0.125C_{\text{Jan2}}$$

$$E_{\text{Jan5}} \approx 0.5(181.18) + 0.25(181.91) + 0.125(184.25) + 0.125(185.64) \approx 182.304$$

Predict: The four weights sum to 1. Will their weighted prices reproduce EMA on 1/5?

```
dates = ["2024-01-02","2024-01-03","2024-01-04","2024-01-05"]
weights = np.array([0.125,0.125,0.25,0.5])
print(f"Four-price weighted sum:{(close.loc[dates]*weights).sum():.4f}")
print(f"Recursive EMA:{ema3['2024-01-05']:.4f}")
```

Four-price weighted sum:182.3037 Recursive EMA:182.3037



The 1/2 price has left the three-observation SMA window but retains weight 0.125 in EMA. The four-price sum and recursion both give 182.3037 USD/share.

EMA retains earlier history and gives greater weight to recent prices. This does not establish predictive ability. Different initialization can produce a different EMA.

Discussion

In EMA3 on 1/5, both 1/2 and 1/3 have weight 0.125. If initialized with the previous trading day's 2023/12/29 Close, what would the 1/2 weight be? Why are the two weights equal here?

Sources: [3], [4], [11].

ISSUE 05

Holding when EMA rises: what does the first trade earn?

Require stock when EMA3 rises, otherwise cash. Decide after the close and assume execution at the next trading day's Close.

Start in cash on 1/2; assume buying and selling one share at Close, without fees, dividends or interest. Find the first EMA rise and the first subsequent decline. Seeing today's rise does not earn today's return.

ema_after_close records stock after execution: True for stock, False for cash. entries marks cash-to-stock buy dates; exits marks stock-to-cash sell dates.

Predict: The 1/8 EMA rise creates a holding requirement. Can the same-day 185.56 be the buy price?

```
ema_up = ema3.diff() > 0
ema_after_close = ema_up.shift(1, fill_value=False)
entries = ema_after_close & ~ema_after_close.shift(1, fill_value=False)
exits = ~ema_after_close & ema_after_close.shift(1, fill_value=False)
ema_buy_date = close.index[entries][0]
ema_sell_date = close.index[exits & (close.index > ema_buy_date)][0]
ema_trade_return = close[ema_sell_date]/close[ema_buy_date]-1
print(f"Buy:{ema_buy_date} at {close[ema_buy_date]:.2f} USD/share")
print(f"Sell:{ema_sell_date} at {close[ema_sell_date]:.2f} USD/share")
print(f"Single-trade price return:{ema_trade_return:+.4%}")
```

Buy:2024-01-09 at 185.14 USD/share Sell:2024-01-17 at 182.68 USD/share Single-trade price return:-1.3287%

EMA rises from 182.3037 to 183.9319 on 1/8; buy at the 1/9 Close, 185.14. On 1/16 it falls from the previous trading day 1/12 value, 185.6982, to 184.6641: the first decline after buying. Sell at the 1/17 Close, 182.68. The holding period is 1/9 to 1/17.

$$\frac{182.68}{185.14} - 1 \approx -1.3287\%$$



An indicator rise only supplies an entry condition. This assumed trade at actual prices loses money; it cannot establish the whole strategy's results.

Discussion

If the 1/8 Close, 185.56, is used as the buy price, which execution assumption changes?

Sources: [4], [11].

ISSUE 06

Does negative MACD mean the price fell that day?

MACD compares shorter- and longer-span EMAs, not today's and yesterday's Closes.

MACD means Moving Average Convergence Divergence. E(12) and E(26) are span12, 26 EMAs, using Issue 3's 1/2 first-price initialization and update. Today's Close weights are α=2/13≈0.1538, 2/27≈0.0741; EMA12 gives recent Closes more weight. Units: USD/share.

$$\mathrm{MACD}_t = E_t^{(12)} - E_t^{(26)}$$

$$E_{\mathrm{Apr25}}^{(12)} \approx 168.67681 + \tfrac{2}{13}(169.89 - 168.67681) \approx 168.86346$$

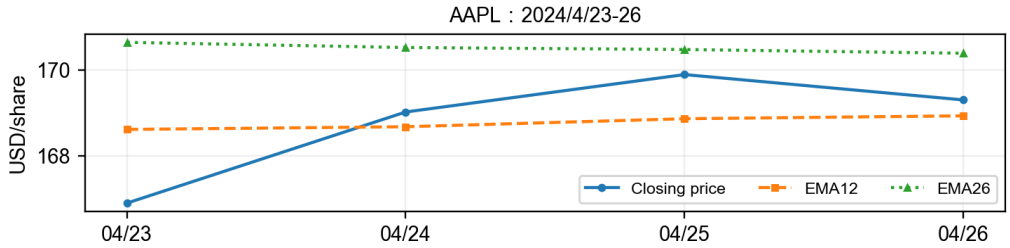
$$E_{\mathrm{Apr25}}^{(26)} \approx 170.51924 + \tfrac{2}{27}(169.89 - 170.51924) \approx 170.47263$$

$$\mathrm{MACD}_{\mathrm{Apr25}} \approx 168.8635 - 170.4726 \approx -1.609$$

Predict: The two EMAs have a negative difference on 4/25. Must that day's Close also fall?

```
ema12 = close.ewm(span=12, adjust=False).mean()
ema26 = close.ewm(span=26, adjust=False).mean()
macd = ema12 - ema26
dates = ["2024-04-24","2024-04-25","2024-04-26"]
pd.DataFrame({'Observation date':dates,'Close (USD/share)':close.loc[dates].round(2),
    'EMA12 (USD/share)':ema12.loc[dates].round(5),'EMA26 (USD/share)':ema26.loc[dates].round(5),
    'MACD (USD/share)':macd.loc[dates].round(5)})
```

Observation date	Close (USD/share)	EMA12 (USD/share)	EMA26 (USD/share)	MACD (USD/share)
2024-04-24	169.02	168.67681	170.51924	-1.84243
2024-04-25	169.89	168.86346	170.47263	-1.60917
2024-04-26	169.3	168.93062	170.38577	-1.45515



MACD is -1.6092 on 4/25, but Close rises from 169.02 on 4/24 to 169.89 USD/share, about +0.5147%. A lower short-span average does not mean a same-day price decline.

A common direction condition requires MACD above zero; 4/25 fails it. Asking whether MACD moves toward zero instead asks whether the gap is narrowing.

Discussion

Using the table, how much does MACD change from 4/24 to 4/26? How does each change in EMA12 and EMA26 contribute?

Sources: [4], [18].

ISSUE 07

How are the signal line and histogram calculated from MACD?

Smooth MACD into a signal line, then subtract today's signal line from today's MACD. Do not subtract Close again.

Start with Close: EMA12 minus EMA26 gives MACD; its 9-period EMA gives the signal line; their difference gives the histogram. S denotes the signal line. Like MACD and the histogram, it is measured in USD/share.

For a 9-period EMA, today's weight is 2/(9+1)=0.2; the remaining 0.8 retains the previous trading day's signal line. Initialize with the first MACD=0 on 1/2, then update daily.

$$S_t = 0.2\,\mathrm{MACD}_t + 0.8S_{t-1}, \qquad \mathrm{Histogram}_t = \mathrm{MACD}_t - S_t$$

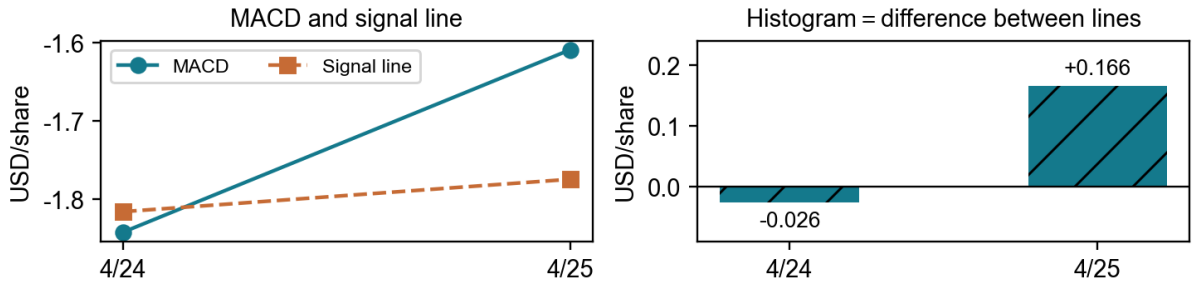
$$S_{\mathrm{Apr25}} \approx 0.2(-1.609) + 0.8(-1.816) \approx -1.775$$

$$\mathrm{Histogram}_{\mathrm{Apr25}} \approx -1.609 - (-1.775) = +0.166$$

Predict: MACD and the signal line are both negative on 4/25. Is their difference positive or negative?

```
signal_line = macd.ewm(span=9, adjust=False).mean()
histogram = macd-signal_line
dates = ["2024-04-24","2024-04-25"]
pd.DataFrame({'Observation date':dates,
    'MACD (USD/share)':macd.loc[dates].round(3),
    'Prior signal line (USD/share)':signal_line.shift(1).loc[dates].round(3),
    'New signal line (USD/share)':signal_line.loc[dates].round(3),
    'Histogram (USD/share)':histogram.loc[dates].round(3)})
```

Observation date	MACD (USD/share)	Prior signal line (USD/share)	New signal line (USD/share)	Histogram (USD/share)
2024-04-24	-1.842	-1.81	-1.816	-0.026
2024-04-25	-1.609	-1.816	-1.775	0.166



On 4/25 MACD is about 0.166 USD/share above its signal line, so the histogram is positive. It changes from -0.026 on 4/24 to positive on 4/25, marking an upward signal-line crossover; it is not a 0.166% stock return.

An upward signal-line crossover qualifies as a candidate entry on 4/25. MACD itself is still below zero, so it fails the different rule MACD>0.

Discussion

For the 4/25 histogram, should MACD subtract that day's signal line -1.775 or the previous day's -1.816? Calculate the histogram.

Sources: [4], [18], [23].

ISSUE 08

Are EMA12 below EMA26 and negative MACD two pieces of evidence?

The conditions are equivalent; requiring both adds no market information.

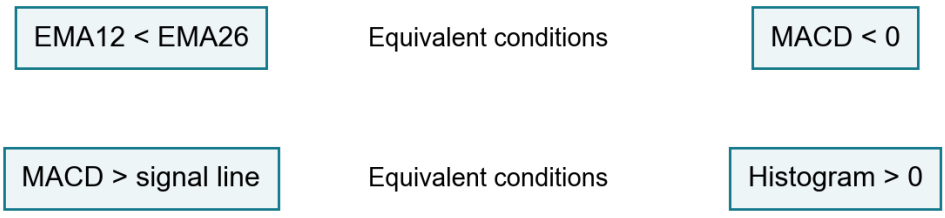
All three values use the same Closes, in USD/share. True means a condition holds that day; False means it does not. These are conditions, not trades or positions. The equivalence follows directly from MACD's definition, not from high correlation:

$$E^{(12)} < E^{(26)} \iff E^{(12)} - E^{(26)} < 0 \iff \text{MACD} < 0$$

Predict: Which two of the three condition columns must give the same True/False result?

```
dates = ["2024-04-24","2024-04-25"]
pd.DataFrame({'Observation date':dates,
    'EMA12 lower':(ema12.loc[dates] < ema26.loc[dates]).values,
    'MACD negative':(macd.loc[dates] < 0).values,
    'Histogram positive':(histogram.loc[dates] > 0).values})
```

Observation date	EMA12 lower	MACD negative	Histogram positive
2024-04-24	True	True	False
2024-04-25	True	True	True



The first two columns are True on 4/24 and 4/25. A positive histogram holds only on 4/25; it answers a different question.

Discussion

If both EMA12 below EMA26 and negative MACD are required, does removing the latter change the selected dates?

Sources: [18], [23].

ISSUE 09

Can short-horizon momentum be positive while longer-horizon momentum is negative?

A price rebound may still be below an earlier starting price. Different horizons can have different signs without contradiction.

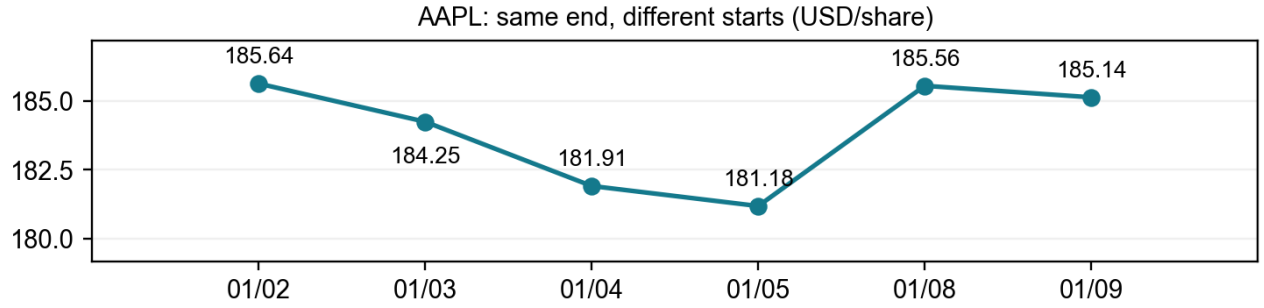
C is Close (USD/share), m is the rate of price change, and h counts intervals between observed trading days. Both horizons end on 1/9; only the starting point changes.

$$m_t^{(h)} = \frac{C_t}{C_{t-h}} - 1, \quad m_{\text{Jan9}}^{(3)} \approx \frac{185.14}{181.91} - 1 \approx 1.7756\%$$

Predict: 185.14 is above 181.91 but below 185.64. Will the two returns have the same sign?

```
momentum3 = close/close.shift(3)-1
momentum5 = close/close.shift(5)-1
starts = ["2024-01-04","2024-01-02"]
pd.DataFrame({'Horizon':['3 intervals','5 intervals'],
    'Start':starts,'Start (USD/share)':close.loc[starts].round(2).values,
    '1/9 end price (USD/share)':[round(close["2024-01-09"],2)]*2,
    'Return':[f"{momentum3['2024-01-09']:+.4%}",
            f"{momentum5['2024-01-09']:+.4%}"]})
```

Horizon	Start	Start (USD/share)	1/9 end price (USD/share)	Return
3 intervals	2024-01-04	181.91	185.14	+1.7756%
5 intervals	2024-01-02	185.64	185.14	-0.2693%



The three-interval return is +1.7756%; the five-interval return is -0.2693%. Under a positive-momentum condition, changing the horizon changes the stock-holding condition.

Discussion

Five-interval momentum is negative on 1/9. For it to turn positive on the next trading day 1/10, which price must the 1/10 Close exceed? Why is the threshold not 185.64 on 1/2?

Sources: [5], [13].

ISSUE 10

Why is volatility nonzero when all three days fall?

Volatility measures each return's distance from the mean, not just its direction.

r is the decimal Close return; its mean is r with a bar, and sample standard deviation is σ with a hat. Close is 185.64 on 1/2 and 184.25, 181.91, 181.18 USD/share on 1/3, 4, 5.

$$\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^{3} (r_i - \bar{r})^2}{3-1}}$$

1/3 return: 184.25÷185.64-1 is about -0.0074876, or -0.7488%. Subtract the three-interval mean: -0.0074876-(-0.0080669) is about +0.0005793.

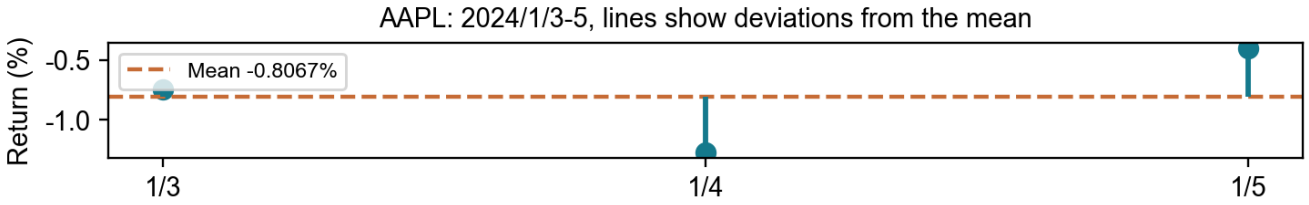
The three-interval mean is about -0.0080669. The three deviations sum to zero, leaving two that can vary independently, so the sample denominator is 3-1=2. Sum squared deviations, then take the square root to return to return units.

$$\hat{\sigma} \approx \sqrt{\frac{0.0005793^2 + (-0.0046332)^2 + 0.0040539^2}{2}} \approx 0.004372 = 0.4372\%$$

Predict: All three returns are negative. Can their deviations from the mean have both signs?

```
daily_return = close.pct_change(fill_method=None)
first_returns = daily_return.iloc[1:4]
mean3 = first_returns.mean()
sd3 = first_returns.std(ddof=1)
pd.DataFrame({'Observation date':first_returns.index,
    'Return (%)':first_returns.map("{:.4%}".format).values,
    'Deviation (decimal)':(first_returns-mean3).map("{:+.7f}".format).values})
```

Observation date	Return (%)	Deviation (decimal)
2024-01-03	-0.7488%	+0.0005793
2024-01-04	-1.2700%	-0.0046332
2024-01-05	-0.4013%	+0.0040539



The three-return standard deviation is about 0.4372%. All returns have the same sign, but different deviations from the mean.

Discussion

Which day contributes the largest squared deviation, and approximately what fraction of the sum of squares?

Sources: [3], [14].

ISSUE 11

Does higher volatility mean a larger same-day loss?

A price rise can increase recent volatility. Volatility measures dispersion, not direction.

Compare 1/5 and 1/8 using three Close returns and sample standard deviation ddof=1; σ with a hat still denotes the return sample standard deviation. The 1/5 window is Issue 10's 1/3, 4, 5; on 1/8 it becomes 1/4, 5, 8, adding +2.4175% on 1/8 and removing -0.7488% on 1/3. Assume 1,000 USD starting value each interval: P&L is 1000 times the decimal return, with standard deviation 1000 times σ with a hat. The amounts are P&L standard deviations, not account values.

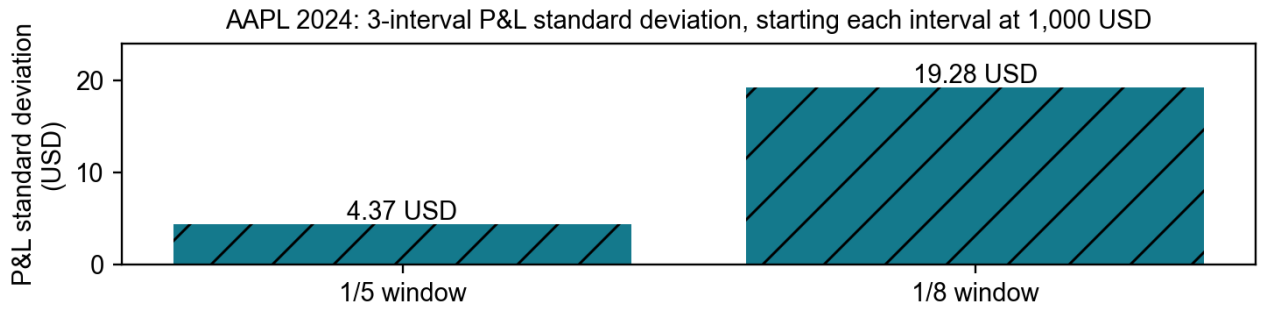
1/5 window: -0.7488%, -1.2700%, -0.4013%; 1/8 window: -1.2700%, -0.4013%, +2.4175%. Remove 1/3 and add 1/8; both windows retain 3 intervals.

$$1000\hat{\sigma} \approx 1000(0.004372) \approx 4.37 \text{ USD}$$

Predict: The 1/8 price jumps 2.4175%. Can volatility be higher than on 1/5?

```
vol3 = daily_return.rolling(3, min_periods=3).std(ddof=1)
dates = ["2024-01-03","2024-01-04","2024-01-05","2024-01-08"]
pd.DataFrame({'Observation date':dates,
    'Daily return':daily_return.loc[dates].map("{:+.4%}".format),
    'Three-interval volatility (%)':vol3.loc[dates].map(lambda x:'Under 3' if pd.isna(x) else f"{x:.4%}"),
    'P&L standard deviation (USD)':(1000*vol3.loc[dates]).map(lambda x:'Under 3' if pd.isna(x) else f"{x:.2f}")})
```

Observation date	Daily return	Three-interval volatility (%)	P&L standard deviation (USD)
2024-01-03	-0.7488%	Under 3	Under 3
2024-01-04	-1.2700%	Under 3	Under 3
2024-01-05	-0.4013%	0.4372%	4.37
2024-01-08	+2.4175%	1.9278%	19.28



The 1/8 standard deviation, 1.9278%, corresponds to 19.28 USD, above 4.37 on 1/5. It is neither a same-day loss nor tomorrow's maximum loss.

Discussion

Can 19.28 USD be interpreted as tomorrow's maximum loss of 19.28 USD?

Sources: [3], [14], [26].

ISSUE 12

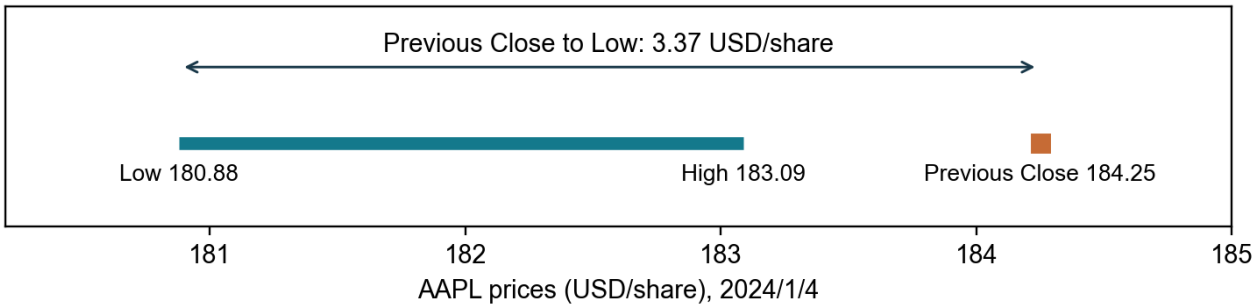
What decline does today's High-Low range miss?

Today's High and Low alone can understate the price change from yesterday's Close.

Assume one AAPL share is already held at the 1/3 Close: its value is 184.25 USD. On 1/4, High is 183.09 and Low is 180.88 USD; even High is below yesterday's Close. H, L denote today's High, Low; C is yesterday's Close. All differences below are USD/share.

$$H - L = 183.09 - 180.88 = 2.21, \qquad C - L = 184.25 - 180.88 = 3.37$$

Observation date	Observation	USD/share
2024/1/3	Closing price	184.25
2024/1/4	High	183.09
2024/1/4	Low	180.88



One share loses 3.37 USD from the previous Close 184.25 to the Low 180.88. The day's 2.21 high-low range does not cover the whole distance.

Today's High-Low is only 2.21 USD, but a share held since yesterday has lost 3.37 USD in value at today's Low. True Range (TR) therefore includes the previous Close. It measures magnitude, not future direction.

Discussion

Which omitted starting point explains the difference between 2.21 and 3.37?

Sources: [19], [21].

ISSUE 13

How can TR of 3.37 USD set a stop distance?

Calculate one day's TR, then use that magnitude for a later holding. It is not a buy signal.

H, L are today's High, Low; C is Close and t-1 is the previous trading day. Units: USD/share. Select the largest nonnegative distance; do not add the three:

$$\mathrm{TR}_t = \max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|)$$

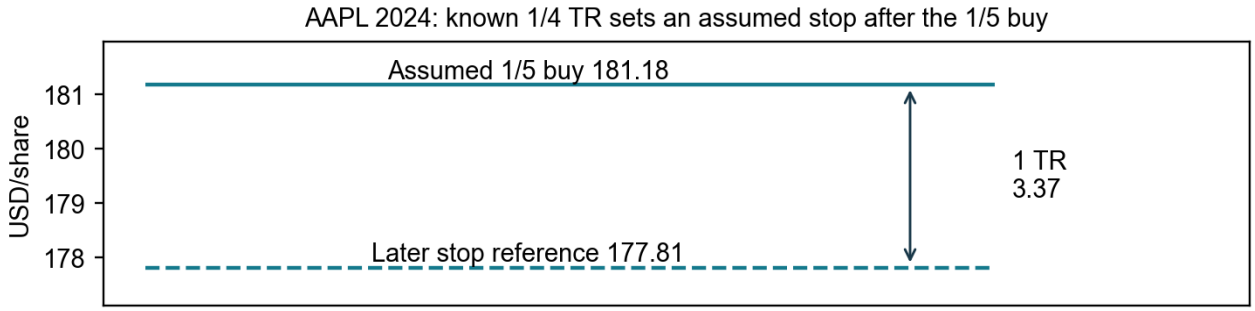
$$\mathrm{TR}_{\mathrm{Jan4}} = \max(2.21, |183.09 - 184.25|, |180.88 - 184.25|) = 3.37$$

TR is about 3.37 after the 1/4 close. Assume buying 10 shares at the actual 1/5 Close, 181.18, and fixing a later stop 1 TR below it: 181.18-3.37=177.81 USD/share. This is a teaching rule; common practice uses smoothed multi-day ATR instead.

Predict: If 10 shares are sold exactly at 177.81, is the loss 3.37 or 33.70 USD?

```
previous_close = close.shift(1)
distances = pd.concat([aapl.High-aapl.Low,
    (aapl.High-previous_close).abs(),
    (aapl.Low-previous_close).abs()],axis=1)
true_range = distances.iloc[1:].max(axis=1)
tr_distance = true_range["2024-01-04"]
tr_buy = close["2024-01-05"]
tr_stop = tr_buy-tr_distance
print(f"TR:{tr_distance:.2f}; trigger:{tr_stop:.2f} USD/share")
print(f"Loss for 10 shares filled exactly at trigger:{10*tr_distance:.2f} USD")
```

TR:3.37; trigger:177.81 USD/share Loss for 10 shares filled exactly at trigger:33.70 USD



The distance is 3.37 per share, or 33.70 USD for 10 shares. Assume a fall to 177.81 requests a sale. Triggering does not guarantee a fill price; gaps, slippage and fees can increase the loss.

Discussion

Keep the buy price and stop unchanged but buy 20 shares. If filled exactly at the trigger, excluding fees, approximately how many USD are lost?

Sources: [19], [21], [27].

ISSUE 14

TR measures a day; how does ATR combine days?

TR measures one day's magnitude. Average True Range (ATR) smooths daily TR values so a distance setting need not depend on a single day.

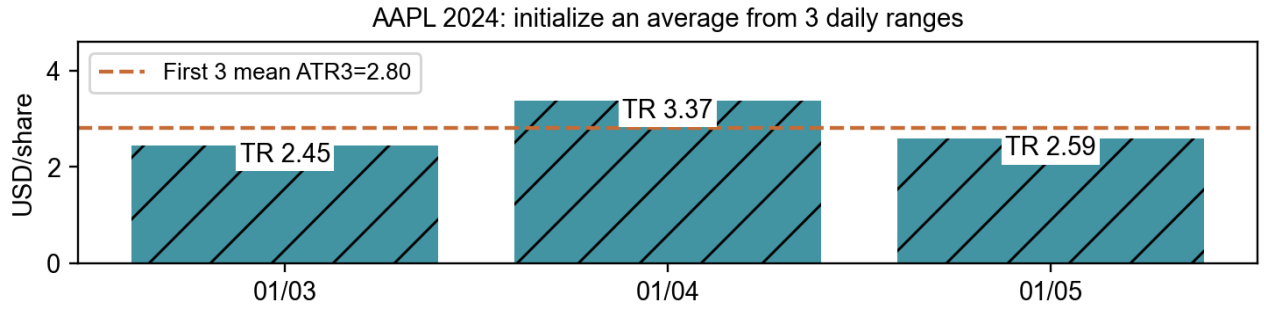
Use 3 periods for illustration; 14 periods is common. All units are USD/share. Initialize ATR with the mean of 3 TR values. Skip 1/2 because its previous Close is unavailable; the first 3 values are 1/3, 4, 5.

$$\mathrm{ATR3}_{\mathrm{Jan5}} = \frac{\mathrm{TR}_{\mathrm{Jan3}} + \mathrm{TR}_{\mathrm{Jan4}} + \mathrm{TR}_{\mathrm{Jan5}}}{3} \approx \frac{2.45 + 3.37 + 2.59}{3} \approx 2.80$$

Predict: Will the mean of 2.45, 3.37, 2.59 lie between their minimum and maximum?

```
seed_tr = true_range.iloc[:3]
atr_seed = seed_tr.mean()
pd.DataFrame({'Observation date':seed_tr.index,
    'TR (USD/share)':seed_tr.round(2).values})
```

Observation date	TR (USD/share)
2024-01-03	2.45
2024-01-04	3.37
2024-01-05	2.59



TR on 1/5 is 2.59. The first ATR3 is about 2.80 because it also includes 1/3 and 1/4, not just 1/5.

ATR14 starts with the first 14 valid TR values: 1/3 through 1/23 here, initially about 3.3507. From the 15th TR, use the next issue's Wilder update, not a freshly calculated simple moving average.

Discussion

On 1/5, TR is 2.59 and ATR3 is about 2.80. Which describes only that day, and which includes the previous two days?

Sources: [19], [21].

ISSUE 15

Today's TR is 4.42; why is ATR only 3.34?

ATR retains part of the previous average and adds today's range; it does not simply become today's TR.

Keep 3 periods: previous ATR is 2.8033 on 1/5; today's TR is 4.42 on 1/8, in USD/share. n is the period count and t denotes today. Wilder retains (n-1)/n of the old average and adds 1/n of today's TR:

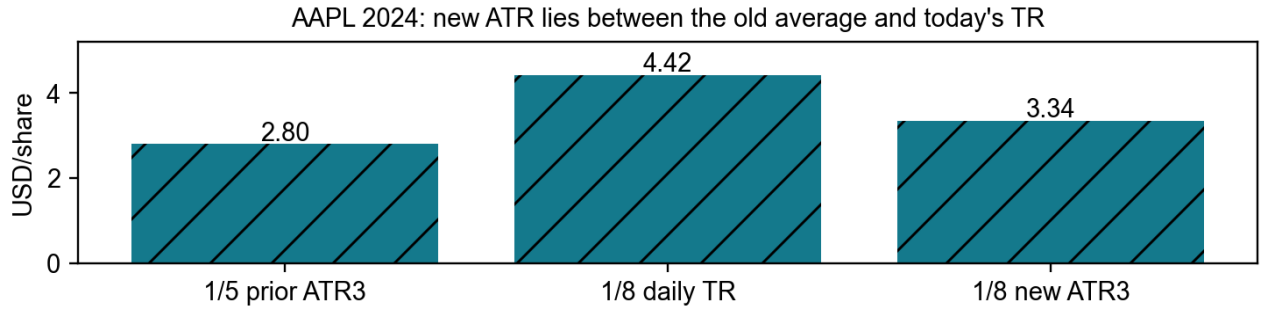
$$\mathrm{ATR}_t = \frac{(n-1)\mathrm{ATR}_{t-1} + \mathrm{TR}_t}{n}$$

$$\mathrm{ATR3}_{\mathrm{Jan8}} \approx \frac{2(2.8033) + 4.42}{3} \approx 3.3422$$

Predict: Today's TR4.42 exceeds prior ATR2.8033. Will the new ATR rise or fall?

```
def wilder_mean(values, n=14):
    result = pd.Series(np.nan,index=values.index)
    result.iloc[n-1] = values.iloc[:n].mean()
    for i in range(n,len(values)):
        result.iloc[i] = ((n-1)*result.iloc[i-1]+values.iloc[i])/n
    return result
atr3 = wilder_mean(true_range, n=3)
atr14 = wilder_mean(true_range)
dates = ["2024-01-05","2024-01-08"]
pd.DataFrame({'Observation date':dates,'Daily TR (USD/share)':true_range.loc[dates].round(2),
    'Prior ATR (USD/share)':['None (initial-value row)',*atr3.shift(1).loc[dates[1:]].round(4)],
    'New ATR3 (USD/share)':atr3.loc[dates].round(4)})
```

Observation date	Daily TR (USD/share)	Prior ATR (USD/share)	New ATR3 (USD/share)
2024-01-05	2.59	None (initial-value row)	2.8033
2024-01-08	4.42	2.8033	3.3422



TR rises from 2.59 to 4.42, but ATR only rises from 2.80 to 3.34. A stop distance based on it is not determined solely by this day's range.

Do not repeatedly average the last 3 values. For 14 periods, use weights 13/14 and 1/14. The code also retains ATR14; later RSI uses the same smoothing function.

Discussion

If the next day's TR is below the current ATR3 of about 3.34, will updated ATR rise or fall?

Sources: [19], [21].

ISSUE 16

How do stops at 1 or 2 ATR differ?

Convert recent price variability into a stop distance. A larger ATR gives a wider distance at the same multiple.

Assume one share was bought at the 1/8 Close, 185.56 USD. ATR3 of about 3.3422 is calculated after the close to set a later stop, not to decide the already-completed buy. C is reference Close in USD/share; k is an assumed multiple, 1 or 2. Fees are excluded.

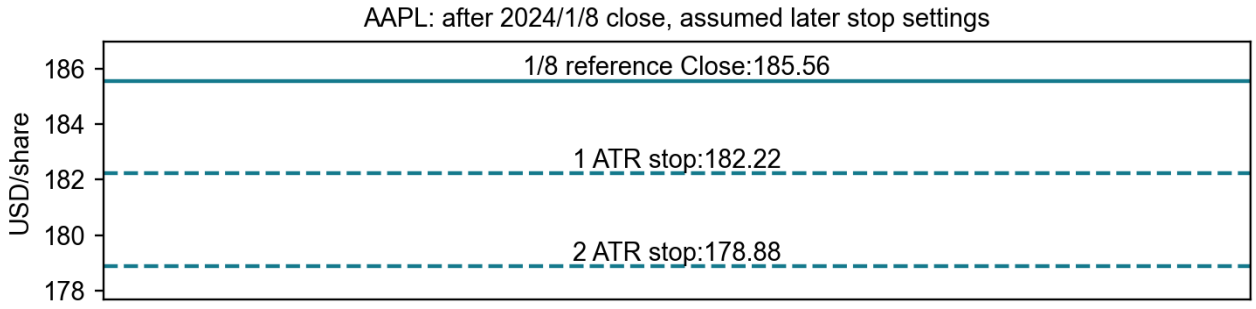
$$\text{Stop trigger} = C - k\,\text{ATR}, \quad 185.56 - 2(3.3422) \approx 178.88$$

Assume a price at or below the trigger activates a market sell order. Triggering does not guarantee execution at that price [27].

Predict: When the multiple changes from 1 to 2, is the trigger nearer to or farther from 185.56?

```
reference_close = close["2024-01-08"]
stop_atr = atr3["2024-01-08"]
pd.DataFrame({'Multiple':[1,2],
    'Distance (USD/share)':[stop_atr,2*stop_atr],
    'Trigger price (USD/share)':[reference_close-stop_atr,reference_close-2*stop_atr]}).round(2)
```

Multiple	Distance (USD/share)	Trigger price (USD/share)
1	3.34	182.22
2	6.68	178.88



The 1 ATR trigger is 182.22; 2 ATR gives 178.88. The latter allows a larger price change and planned loss. The multiples are not optimized; ATR is not a cap on tomorrow's decline.

Discussion

With the same buy price, 185.56, and fees excluded, what is the approximate one-share loss if filled exactly at each trigger?

Sources: [21], [27].

ISSUE 17

How does RSI convert daily price changes to 0 to 100?

RSI compares smoothed gain and loss amounts. U-D is only the day's price change, not RSI.

C is Close. U and D record daily gain and loss amounts, both nonnegative and measured in USD/share. Bars denote their separate Wilder-smoothed averages, not averages over only rising or falling days.

$$U_t = \max(C_t - C_{t-1}, 0), \qquad D_t = \max(C_{t-1} - C_t, 0)$$

$$\bar{U}_{\text{first}} = \frac{\sum_{i=1}^{14} U_i}{14}, \qquad \bar{D}_{\text{first}} = \frac{\sum_{i=1}^{14} D_i}{14}$$

$$\bar{U}_t = \frac{13\bar{U}_{t-1} + U_t}{14}, \qquad \bar{D}_t = \frac{13\bar{D}_{t-1} + D_t}{14}$$

$$\text{RSI}_t = 100\frac{\bar{U}_t}{\bar{U}_t + \bar{D}_t}$$

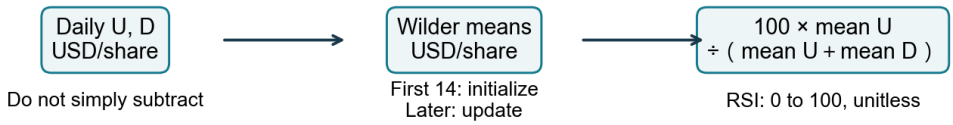
Separate daily gains and losses; divide each sum over the first 14 observations by 14; then use 13 times the previous average plus 1 times today's value, divided by 14; substitute into RSI. A 0 denominator remains undefined here. RSI has no USD unit.

Predict: On 1/8 Close rises from 181.18 to 185.56. Should we record U=4.38 or D=4.38?

```
change = close.diff().iloc[1:]
gains, losses = change.clip(lower=0), (-change).clip(lower=0)
dates = ["2024-01-03","2024-01-08"]
pd.DataFrame({'Observation date':dates,
    'Prior Close (USD/share)':close.shift(1).loc[dates].round(2),
    'Close (USD/share)':close.loc[dates].round(2),
    'U (USD/share)':gains.loc[dates].round(2),
    'D (USD/share)':losses.loc[dates].round(2)})
```

Observation date	Prior Close (USD/share)	Close (USD/share)	U (USD/share)	D (USD/share)
2024-01-03	185.64	184.25	0.0	1.39
2024-01-08	181.18	185.56	4.38	0.0

1/3: U=0, D=1.39; 1/8: U=4.38, D=0. Issues 18 to 20 calculate initial averages, daily updates and RSI in order.



RSI compares the relative size of the two averages, not today's USD gain or percentage profit.

Discussion

If today's U=0 and D>0, are those two values alone sufficient to calculate today's RSI?

Sources: [20].

ISSUE 18

How do 14 changes initialize RSI's two averages?

List all 14 gains and losses. Both averages divide by 14, retaining zeros on days without a gain or loss.

AAPL, 2024/1/3-23: 14 Close changes require the previous Close on 1/2. U and D are gain and loss amounts; table values and their averages are all in USD/share.

Predict: With days where U=0, is mean U's denominator still 14 or the count of rising days?

```
gain_seed, loss_seed = gains.iloc[:14], losses.iloc[:14]
initial_up, initial_down = gain_seed.mean(), loss_seed.mean()
pd.DataFrame({'Observation date':gain_seed.index,
    'U (USD/share)':gain_seed.round(2).values,
    'D (USD/share)':loss_seed.round(2).values})
```

Observation date	U (USD/share)	D (USD/share)
2024-01-03	0.0	1.39
2024-01-04	0.0	2.34
2024-01-05	0.0	0.73
2024-01-08	4.38	0.0
2024-01-09	0.0	0.42
2024-01-10	1.05	0.0
2024-01-11	0.0	0.6
2024-01-12	0.33	0.0
2024-01-16	0.0	2.29
2024-01-17	0.0	0.95
2024-01-18	5.95	0.0
2024-01-19	2.93	0.0
2024-01-22	2.33	0.0
2024-01-23	1.29	0.0

$$\bar{U}_{\text{Jan23}} \approx \frac{18.26}{14} \approx 1.3043, \qquad \bar{D}_{\text{Jan23}} \approx \frac{8.72}{14} \approx 0.6229$$

The table's U sum is about 18.26 and D sum about 8.72; divide each by 14. Do not average only the last three displayed rows, or remove zeros and divide by rising/falling day counts.

Discussion

If 1/3 with U=0 is removed and the denominator becomes 13, is this still the original RSI14 initialization?

Sources: [19], [20].

ISSUE 19

How are yesterday's mean U and D updated today?

The right-hand columns are neither daily changes nor ordinary averages of the displayed days. Each row updates the previous trading day's averages.

A 14-period Wilder average retains 13/14 of the previous average and adds 1/14 of today's value. The 1/23 initialization uses the preceding issue's 14 observations. All amounts are USD/share; extra digits in prior means let the four-decimal updates be checked directly.

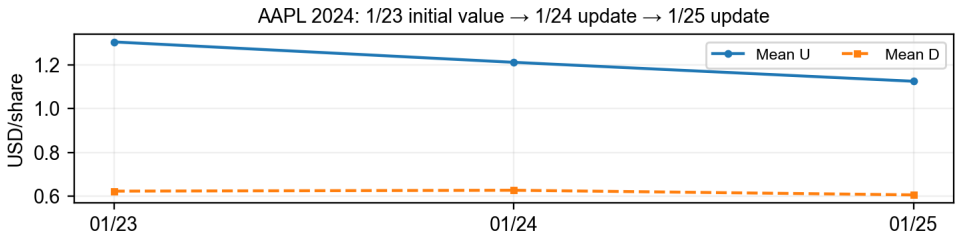
$$\bar{U}_{\text{Jan24}} \approx \frac{13(1.304286)+0}{14} \approx 1.2111, \quad \bar{D}_{\text{Jan24}} \approx \frac{13(0.622858)+0.68}{14} \approx 0.6269$$

$$\bar{U}_{\text{Jan25}} \approx \frac{13(1.211123)+0}{14} \approx 1.1246, \quad \bar{D}_{\text{Jan25}} \approx \frac{13(0.626939)+0.33}{14} \approx 0.6057$$

Predict: With no gain on 1/24, does new mean U become 0 immediately or retain some earlier gains?

```
up, down = wilder_mean(gains), wilder_mean(losses)
dates = ["2024-01-24","2024-01-25"]
pd.DataFrame({'Observation date':dates,
    'U (USD/share)':gains.loc[dates].round(2),
    'D (USD/share)':losses.loc[dates].round(2),
    'Prior mean U (USD/share)':up.shift(1).loc[dates].round(6),
    'Prior mean D (USD/share)':down.shift(1).loc[dates].round(6),
    'New mean U (USD/share)':up.loc[dates].round(4),
    'New mean D (USD/share)':down.loc[dates].round(4)})
```

Observation date	U (USD/share)	D (USD/share)	Prior mean U (USD/share)	Prior mean D (USD/share)	New mean U (USD/share)	New mean D (USD/share)
2024-01-24	0.0	0.68	1.304286	0.622858	1.2111	0.6269
2024-01-25	0.0	0.33	1.211123	0.626939	1.1246	0.6057



U=0 on 1/24, but mean U remains 1.2111. The 1/25 loss, 0.33, is below prior mean D of about 0.626939, so new mean D falls to 0.6057.

Close is 195.18 on 1/23, 194.50 on 1/24 and 194.17 on 1/25. Today's D comes from adjacent Closes; calculate RSI after updating both averages.

Discussion

For mean U on 1/25, should 13 multiply 1.304286 from 1/23 or 1.211123 from 1/24? Why?

Sources: [19], [20].

ISSUE 20

What does RSI 65.89 measure?

Use the preceding issue's two smoothed means in the ratio. A greater gain share gives higher RSI; it is not a holding return.

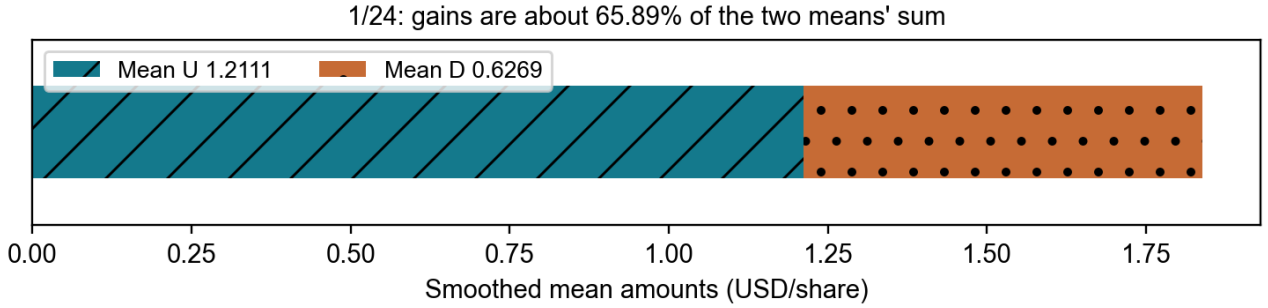
Mean U and mean D are both in USD/share. Their sum is the denominator; the USD units cancel, giving RSI from 0 to 100. A zero sum remains undefined here.

$$\mathrm{RSI}_{\mathrm{Jan24}} \approx 100\frac{1.2111}{1.2111 + 0.6269} \approx 65.89$$

Predict: Close falls 0.3484% on 1/24 but mean U still exceeds mean D. Can RSI exceed 50?

```
total = up+down
rsi14 = 100*up/total.where(total > 0)
dates = ["2024-01-23","2024-01-24","2024-01-25"]
pd.DataFrame({'Observation date':dates,
    'Mean U (USD/share)':up.loc[dates].round(4),
    'Mean D (USD/share)':down.loc[dates].round(4),
    'RSI14 (unitless)':rsi14.loc[dates].round(2),
    'Daily price return':daily_return.loc[dates].map("{:+.4%}".format)})
```

Observation date	Mean U (USD/share)	Mean D (USD/share)	RSI14 (unitless)	Daily price return
2024-01-23	1.3043	0.6229	67.68	+0.6653%
2024-01-24	1.2111	0.6269	65.89	-0.3484%
2024-01-25	1.1246	0.6057	64.99	-0.1697%



RSI above 50 on 1/24 means only that mean U exceeds mean D. It does not imply a same-day rise, a 65.89% profit or a future rise probability.

Sources often first calculate RS=mean U÷mean D, then RSI=100-100/(1+RS). For mean D>0 this equals the displayed formula. Using the share directly makes the two averages' roles visible.

Discussion

If mean U and mean D are equal and both positive, what is RSI? Does it imply an investment return of the same proportion?

Sources: [20].

ISSUE 21

Does RSI above 70 guarantee a fall next trading day?

70 is a common overbought threshold, not a guarantee of reversal. Check all qualifying dates in this sample, not just one counterexample.

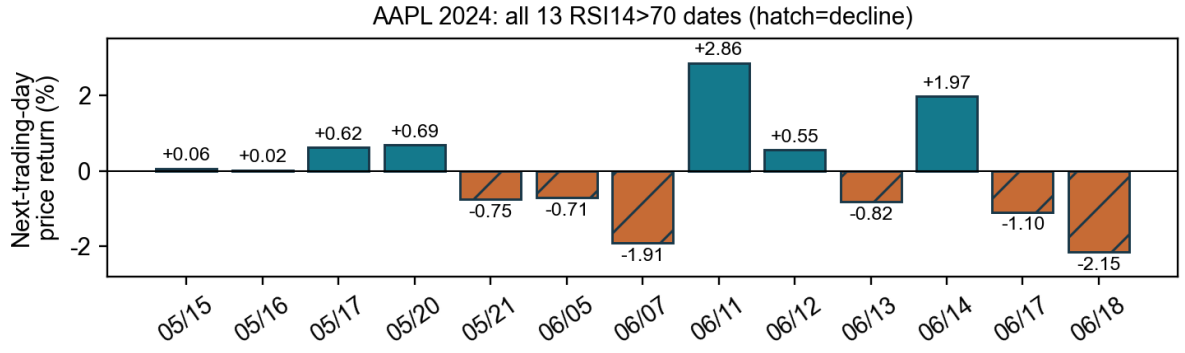
AAPL, 2024/1/2-6/28; use RSI14 from Issues 18-20. Include only defined RSI with a next Close; do not fill with 0. C is Close (USD/share), and t+1 the next trading day. Close is 207.15 on 6/11 and 213.07 on 6/12.

$$r = \frac{C_{t+1}}{C_t} - 1; \qquad \frac{213.07}{207.15} - 1 \approx +2.8578\%$$

Predict: Is every next-trading-day return after RSI>70 in this sample negative?

```
events = pd.DataFrame({"RSI":rsi14.reindex(close.index),
    'Closing price':close, 'Next trading date':pd.Series(close.index,index=close.index).shift(-1),
    'Next Close':close.shift(-1)}).dropna()
events['Next-day return'] = events['Next Close']/events['Closing price']-1
high_rsi = events.loc[events["RSI"] > 70]
shown = high_rsi.loc[["2024-05-15","2024-06-07","2024-06-11","2024-06-18"]].copy()
shown.insert(0,'Observation date',shown.index)
shown[["RSI",'Closing price','Next Close']] = shown[["RSI",'Closing price','Next Close']].round(2)
shown['Next-day return'] = shown['Next-day return'].map("{:+.4%}".format)
shown
```

Observation date	RSI	Closing price	Next trading date	Next Close	Next-day return
2024-05-15	70.83	189.72	2024-05-16	189.84	+0.0632%
2024-06-07	70.92	196.89	2024-06-10	193.12	-1.9148%
2024-06-11	75.22	207.15	2024-06-12	213.07	+2.8578%
2024-06-18	73.57	214.29	2024-06-20	209.68	-2.1513%



Of 13 events, 7 rise and 6 fall: 6/11→12 rises 2.8578%, and 6/7→10 falls 1.9148%. The trading day after 6/18 is 6/20. Consecutive high-RSI dates are not independent trials; 7/13 is this sample's proportion, not a future rise probability.

A no-new-buy rule for RSI>70 excludes all 13 events: it may avoid declines but also miss further rises. Signals form after the close; the plotted changes are not returns obtainable by a later order. See Issue 5 for execution timing.

Discussion

Compare 6/7 and 6/11: why does RSI>70 not directly imply next-trading-day return<0?

Sources: [20].

ISSUE 22

Why can price fall when volume rises 58%?

A volume change concerns shares traded, not a holding return; the two are not interchangeable.

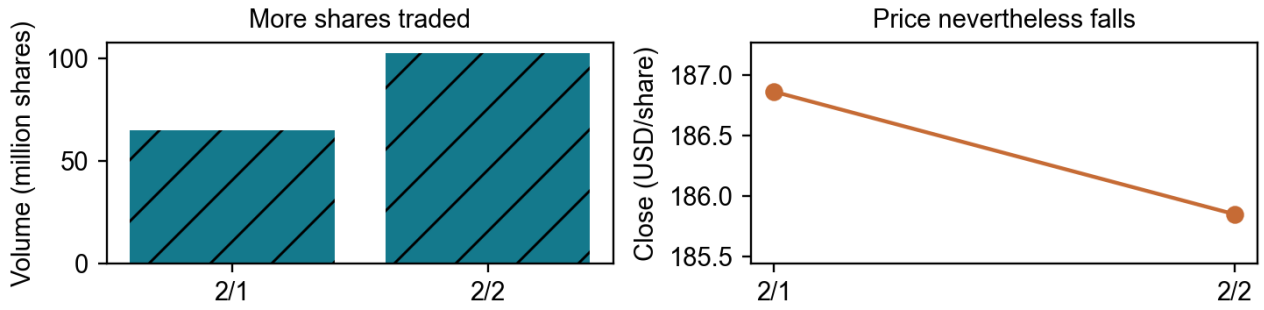
V is daily volume (shares), C is Close (USD/share), g is volume growth, and r is price return. Both compare 2/2 with the previous trading day 2/1, using their 2/1 values as denominators.

$$g = \frac{V_2}{V_1} - 1 = \frac{102551700}{64885400} - 1 \approx 58.05\%, \quad r \approx \frac{185.85}{186.86} - 1 \approx -0.54\%$$

Predict: Does more trading on 2/2 mean shareholders earn 58.05%?

```
volume = aapl["Volume"]
volume_change = volume/volume.shift(1)-1
dates = ["2024-02-01","2024-02-02"]
pd.DataFrame({'Observation date':dates,'Close (USD/share)':close.loc[dates].round(2),
    'Volume (shares)':volume.loc[dates].values,
    'Price return':["-",f"{daily_return['2024-02-02']:+.2%}"],
    'Volume change':["-",f"{volume_change['2024-02-02']:+.2%}"]})
```

Observation date	Close (USD/share)	Volume (shares)	Price return	Volume change
2024-02-01	186.86	64885400	-	-
2024-02-02	185.85	102551700	-0.54%	+58.05%



Volume on 2/2 is 102.55 million shares, 58.05% above 64.89 million; Close nevertheless falls 1.01 USD.

Higher volume alone does not measure buyers' bullishness. Every trade has a buyer and seller; daily total volume does not identify the initiating side.

Discussion

Is the denominator of 58.05% yesterday's stock price or shares traded?

Sources: [15], [24].

ISSUE 23

Can volume exceed yesterday's but remain below its usual level?

Identify the comparison baseline. After a low-volume day, today's increase may still be below the previous 20-observation average.

V is shares traded; M is average volume over the previous 20 observations, excluding today. On 2/7, M uses 20 trading days from 1/9 to 2/6: 1,133,676,500 shares÷20=56,683,825 shares. V/M is a multiple; 1 means average volume, not a 100% investment return.

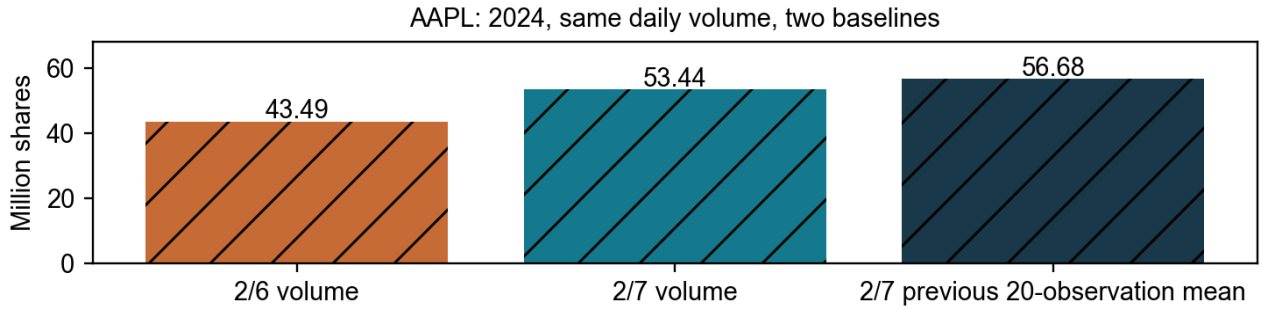
The 2/6 row instead uses 20 observations from 1/8 to 2/5: 1,149,330,200 shares÷20=57,466,510 shares. Each day uses only its preceding 20 observations; the window moves with the date.

$$\frac{V_{\text{Feb7}}}{M_{\text{Feb7}}} = \frac{53439000}{56683825} \approx 0.943, \quad \frac{53439000}{43490800} - 1 \approx 22.87\%$$

Predict: Are 53,439,000 shares on 2/7 above both yesterday's volume and the previous 20-observation mean?

```
prior20 = volume.shift(1).rolling(20,min_periods=20).mean()
volume_multiple = volume/prior20
dates = ["2024-02-06","2024-02-07"]
pd.DataFrame({'Observation date':dates,'Volume (shares)':volume.loc[dates],
    'Prior 20 mean (shares)':prior20.loc[dates].round(0),
    'Multiple (times)':volume_multiple.loc[dates].round(3)})
```

Observation date	Volume (shares)	Prior 20 mean (shares)	Multiple (times)
2024-02-06	43490800	57466510.0	0.757
2024-02-07	53439000	56683825.0	0.943



Volume on 2/7 rises 22.87% from yesterday but is only 0.943 times average volume. A rising-price condition with above-average volume is not met.

Close rises from 189.30 on 2/6 to 189.41 USD/share on 2/7, about 0.06%. The original condition excludes this date because volume is below its average.

Discussion

If higher volume instead means above yesterday's volume, does 2/7 qualify?

Sources: [3], [15], [24].

ISSUE 24

Does an average being above another mean a golden cross?

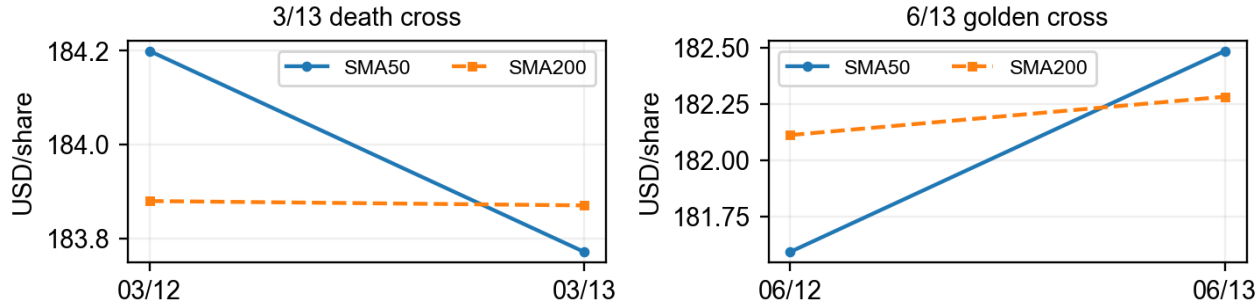
A crossover changes the ordering between two days; being above describes only today's state.

SMA50 and SMA200 average 50 and 200 Closes, in USD/share. A golden cross occurs when yesterday's 50 line is not above the 200 line, but today's is above; a death cross reverses this. Obtain earlier history to calculate 200 observations.

Predict: The 50 line is higher on 6/13. Which other day determines whether it just crossed?

```
long_history = yf.Ticker("AAPL").history(start="2023-01-01",
    end="2024-07-01",interval="1d",auto_adjust=False,actions=True)
long_history.index = long_history.index.strftime("%Y-%m-%d")
long_close = long_history["Close"]
assert np.allclose(long_close.loc[close.index],close,rtol=0,atol=1e-10)
sma50,sma200 = long_close.rolling(50).mean(),long_close.rolling(200).mean()
golden = (sma50>sma200) & (sma50.shift(1)<=sma200.shift(1))
death = (sma50<sma200) & (sma50.shift(1)>=sma200.shift(1))
dates=["2024-03-12","2024-03-13","2024-06-12","2024-06-13"]
pd.DataFrame({'Observation date':dates,'SMA50 (USD/share)':sma50.loc[dates].round(4),
    'SMA200 (USD/share)':sma200.loc[dates].round(4)})
```

Observation date	SMA50 (USD/share)	SMA200 (USD/share)
2024-03-12	184.1992	183.8796
2024-03-13	183.7712	183.8703
2024-06-12	181.5936	182.1121
2024-06-13	182.4854	182.2824



Death cross: 3/12 SMA50 184.1992>183.8796, then 3/13 183.7712<183.8703. Golden cross: 6/12 181.5936<182.1121, then 6/13 182.4854>182.2824. These are candidate conditions, not guarantees of subsequent price direction.

Discussion

After 6/13, while the 50 line stays above the 200 line, is every day a new golden cross?

Sources: [3], [24].

ISSUE 25

Which dates require stock when both MACD and RSI must qualify?

Explicit indicator conditions can be calculated and checked more directly than a vague claim of a favorable trend.

Here stock is required only if histogram>0 and RSI14<70; otherwise cash. AND requires both; OR requires at least one. In the AND column, True requires stock after that day's close; False means at least one condition fails. These are conditions, not fills. Incomplete inputs remain nan, not False or zero.

Predict: On 6/11, histogram 0.5214>0 but RSI is 75.22. Does AND hold?

```
dates = ["2024-05-06","2024-06-07","2024-06-10","2024-06-11"]
valid = histogram.notna() & rsi14.reindex(close.index).notna()
combined = ((histogram>0) & (rsi14.reindex(close.index)<70)).where(valid)
pd.DataFrame({'Observation date':dates,'Histogram (USD/share)':histogram.loc[dates].round(4),
    'RSI (unitless)':rsi14.loc[dates].round(2),
    'AND holds':combined.loc[dates].values})
```

Observation date	Histogram (USD/share)	RSI (unitless)	AND holds
2024-05-06	1.6806	63.79	True
2024-06-07	0.0507	70.92	False
2024-06-10	-0.1925	60.18	False
2024-06-11	0.5214	75.22	False

Date	Histogram>0	RSI<70	AND requirement
05/06	Met	Met	Stock
06/07	Met	Not met	Cash
06/10	Not met	Met	Cash
06/11	Met	Not met	Cash

Both hold on 5/6, requiring stock. RSI fails on 6/7 and 11; the histogram fails on 6/10.

This is only a candidate trading rule; execution and account amounts are not yet specified. ATR may supply distance, but not direction by itself.

Discussion

If only AND changes to OR on 6/11, does the stock requirement change?

Sources: [18], [20], [24].

ISSUE 26

Can today's final-Close SMA be calculated at 15:00?

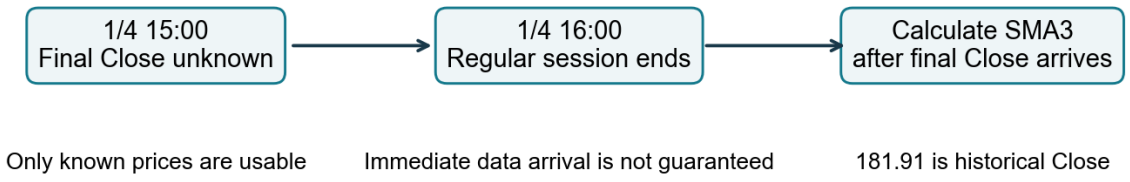
A price labeled with today's date is not necessarily known throughout that day.

Suppose SMA3 is needed at 15:00 New York time on 1/4 using three final Closes. Regular trading ends at 16:00; the previous two Closes have arrived, but final 1/4 Close does not yet exist.

Close date	Known at 15:00 (USD/share)	Usable for original rule
2024/1/2	185.64	Yes
2024/1/3	184.25	Yes
2024/1/4	Not yet known	No

$$\mathrm{SMA3}_{\mathrm{Jan4}} = \frac{185.64 + 184.25 + 181.91}{3} \approx 183.9333$$

181.91 is the 1/4 Close in the historical file; it cannot be backfilled as known at 15:00. Intraday prices can support another rule, but its changed inputs must be declared.



Correct arithmetic cannot replace unavailable inputs. A daily historical file does not establish the actual historical receipt time.

Discussion

If 184.25 from 1/3 is used as the 1/4 Close, is this still the original SMA3?

Sources: [1], [2].

ISSUE 27

Why is a closer centered average unavailable that day?

It includes the following day's price under today's label. A closer line does not mean earlier information.

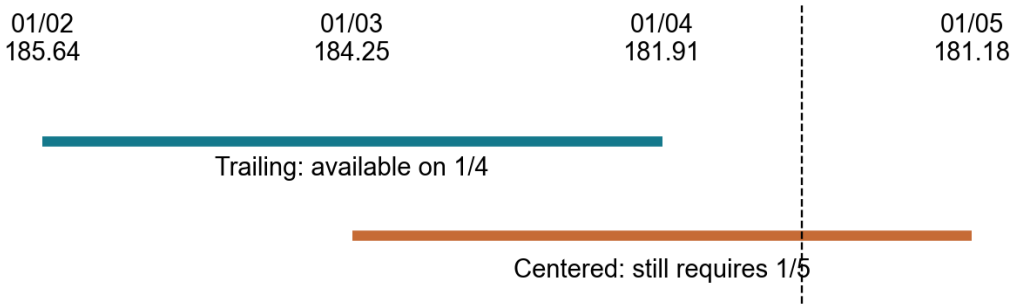
The trailing SMA on 1/4 uses 1/2, 3, 4; the centered mean uses 1/3, 4, 5. C is Close in USD/share; both divide by 3.

$$\text{Centered}_{\text{Jan4}} = \frac{184.25 + 181.91 + 181.18}{3} \approx 182.4467$$

Predict: Without prices from 1/5 onward, can the centered mean on 1/4 be calculated?

```
centered3 = close.rolling(3,center=True,min_periods=3).mean()
prefix = close.loc[:"2024-01-04"]
print(f"1/4 trailing:{sma3['2024-01-04']:.4f}")
print(f"Retrospective 1/4 centered:{centered3['2024-01-04']:.4f}")
print('Centered as of 1/4:',prefix.rolling(3,center=True).mean().iloc[-1])
```

1/4 trailing:183.9333 Retrospective 1/4 centered:182.4467 Centered as of 1/4: nan



Trailing 183.9333 is calculable; centered 182.4467 must wait for 1/5. nan means that as of 1/4, the 1/5 price is missing, so the centered value is unavailable.

Discussion

Does the centered value being closer to the 1/4 Close, 181.91, establish predictive ability?

Sources: [3].

References and Reproduction

The following sources support calculation definitions, common uses and data dates.

[1] yfinance: historical prices

[2] Nasdaq: regular trading hours

[3] pandas: rolling windows

[4] pandas: exponential weighting

[5] pandas: shifting rows

[10] Fidelity: SMA

[11] Fidelity: EMA

[12] Lo, Mamaysky and Wang: technical analysis research

[13] Fidelity: rate of price change

[14] NIST: standard deviation

[15] Nasdaq: trading volume

[17] Yahoo Finance: AAPL historical prices

[18] Fidelity: MACD

[19] StockCharts: ATR calculation and initialization

[20] StockCharts: RSI calculation and use

[21] Fidelity: TR and ATR

[23] TradingView: MACD histogram

[24] Fidelity: technical analysis and common uses

[26] Fidelity: historical volatility and risk

[27] Investor.gov: stop triggers do not guarantee fill prices

Reproduce with retained Yahoo/yfinance data, without a new download. Main AAPL dates are 2024/1/2-6/28; SMA200 uses history from 2023, with matching overlapping Closes. Indicator calculations can be checked; this does not verify historical data receipt or actual fills.